

Early versus late amniotomy for labor induction: A randomized trial

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OBJECTIVE: Our purpose was to determine the impact of early and late amniotomy on labor induction with continuous oxytocin infusion at term.

STUDY DESIGN: A total of 209 women admitted for labor induction were randomized to early or late amniotomy. The early amniotomy group ($n = 106$) had membranes ruptured as soon as it was deemed safe and feasible. The late amniotomy group ($n = 103$) had membrane rupture performed at ≥ 5 cm dilatation. The first 103 women received a continuous oxytocin infusion with incremental adjustments at 60-minute intervals as required. The next 106 women had adjustments every 30 minutes as required. Statistical analysis was confined to concurrent groups.

RESULTS: Early amniotomy was associated with shorter labor (13.3 vs 17.8 hours, $p = 0.001$), chorioamnionitis (22.6% vs 6.8%, $p = 0.002$), and significant fetal umbilical cord compression (12.3% vs 2.9%, $p = 0.017$). The benefit regarding shortening of labor was limited to women having oxytocin increments every 30 minutes as required (13.3 vs 17.8 hours, $p = 0.001$). Alternatively, the increase in chorioamnionitis was confined to the 60-minute group (39% vs 11%, $p < 0.001$), which also demonstrated a trend toward increased moderate and severe variable decelerations (19.6% vs 6.4%, $p = 0.08$).

CONCLUSIONS: When a protocol of 60-minute increments in oxytocin infusion rate is desired, amniotomy should be performed late in labor to reduce chorioamnionitis and significant umbilical cord compression. Alternatively, if early amniotomy is necessary, oxytocin should be adjusted every 30 minutes as tolerated. (AM J OBSTET GYNECOL 1995;173:1321-5.)

Key words: Labor, induction, amniotomy, amniorrhexis

Labor induction continues to be one of the most commonly performed procedures in the United States. In spite of its frequency, considerable controversy exists regarding the optimal timing of oxytocin increments, dosage of oxytocin, and whether membrane rupture early in the induction process offers significant advantage. In vivo studies have demonstrated oxytocin to reach steady state after approximately 40 minutes of continuous infusion.¹⁻² Nevertheless, current literature continues to recommend oxytocin increments over a wide time range (every 20 to 60 minutes).³⁻⁵ Amniotomy has been demonstrated to increase prostaglandin levels in amniotic fluid and maternal plasma.⁶⁻⁸ It is frequently used in combination with continuous oxytocin infusion for labor induction. We have previously proposed oxytocin increments at 60 minutes to be a safe and effective technique for labor induction and have found such a

protocol to be associated with lower incidences of uterine hyperstimulation and abnormal fetal heart rate (FHR) tracings compared with increments every 20 minutes as required.³ However, no study to date has evaluated the utility of amniotomy at different oxytocin dosage intervals.

The purpose of this randomized prospective trial was to evaluate the impact of early and late amniotomy on labor induction in women having incremental increases of oxytocin at 30- or 60-minute intervals.

Material and methods

This study was performed at the E.H. Crump Women's Hospital and was approved by the Institutional Review Board of the University of Tennessee. Women admitted for induction were considered for participation in this randomized clinical trial if they were between 37 and 42 weeks' gestation with intact membranes and demonstrated no fetal or maternal contraindications to a trial of labor. Patients known to have intrauterine infections or other indications for antimicrobial treatment and those having prior cervical ripening with prostaglandins were excluded from participation.

Eligible and consenting women were then random-

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Table I. Demographic and clinical information

	Early amniotomy (n = 106)	Late amniotomy (n = 103)
Maternal age (yr)	23.6 (6.3)	25.3 (6.4)
Race (% black)	75.3	75.8
Gestational age (wk)	40.0 (1.6)	39.8 (1.7)
Gravidity*	2	2
Parity*	1	1
Nulliparity (%)	44.6	55.4
Bishop score*	5	5
Bishop score < 6	50.6	56.5
Cervical dilatation (cm)*	1.5	2
Cervical dilatation < 3 cm (%)	75.7	81.1
Estimated fetal weight (gm)	3245 (724)	3223 (664)

*Median. Numbers in parentheses represent SD.

ized to either "early" or "late" amniotomy by a computer-generated random number scheme. Blinding of physicians and patients before randomization was accomplished with sequentially numbered, sealed opaque envelopes that contained preprinted orders for oxytocin and amniotomy regimens. The early amniotomy group underwent either immediate amniotomy or amniorrhexis as soon as it deemed safe and feasible by the attending physician. The late amniotomy group was not to undergo amniotomy until 5 cm dilatation, unless the FHR tracing suggested the possibility of fetal compromise. The first 103 women received continuous oxytocin infusion with incremental adjustments at 60-minute intervals as required to establish progressive labor. The next 106 women had oxytocin adjustments every 30 minutes as required.

Each patient had a digital cervical examination performed for evaluation of Bishop's score before enrollment. Continuous FHR and uterine contraction monitoring were performed until delivery. Oxytocin was started at 1 mU/min and increased subsequently to 2, 4, 8, 16, and 32 mU/min, as necessary. An adequate contraction pattern was defined as three to five contractions in a 10-minute period, with each contraction lasting 45 to 75 seconds. Oxytocin was decreased or discontinued whenever possible in the active phase of labor and was discontinued during insertion of epidural cannulas. Oxytocin was discontinued or decreased at the discretion of the attending physician or supervising nurse for evidence of uterine hyperstimulation (more than five contractions in 10 minutes, contractions lasting >90 seconds) or evidence of fetal compromise including repetitive late-onset decelerations, recurrent severe variable decelerations, or sustained fetal tachycardia with decreased beat-to-beat variability. Amnioinfusion was routinely used for patients with either thick meconium or moderate to severe variable decelerations, unless delivery was imminent or emergency delivery was required.

Maternal demographic factors, clinical characteristics, ultrasonographic estimation of fetal weight, and

indications for labor induction were evaluated at admission. Labor course including latency information, uterine hyperstimulation requiring oxytocin adjustment, abnormal FHR patterns, number of digital examinations, epidural anesthesia, and chorioamnionitis were characterized. To remove potential bias related to fetal distress and dystocia, labor latency data were calculated only for those women delivered vaginally.

Statistical analysis included descriptive statistics, Fisher's exact test for bivariate data, and the two-tailed Student *t* test for continuous data. A *p* value ≤ 0.05 was considered statistically significant. Study participants were evaluated within their assigned study groups regardless of induction outcome. Subgroup analysis was confined to concurrent groups.

Results

A total of 209 women were enrolled in the trial, with 106 and 103 assigned to early or late amniotomy, respectively. Relevant demographic characteristics and clinical findings are outlined in Table I. The study groups were similar in maternal age, race, gestational age, gravidity, parity, Bishop score, cervical dilatation, and estimated fetal weight. Indications for induction were also similar between the early and late amniotomy groups, with postdates pregnancy (42 completed weeks' gestation) being the most frequent indication (47% vs 41%), followed by preeclampsia (29% vs 28%), fetal growth restriction (10% vs 5%), maternal medical disease (4% vs 10%), and other (9% vs 15%).

The median cervical dilatation at amniorrhexis was 2.0 and 5.0 cm, respectively, in the early and late amniotomy groups. The median intervals from enrollment to membrane rupture were 0.28 and 9.9 hours. Both early and late amniotomy groups underwent a similar number of digital pelvic examinations (median: six in each). Women in the 60-minute increment arm received a median of seven digital examinations versus five in the 30-minute arm (not significant).

Intrapartum characteristics are described in Table II. Severe variable decelerations were seen more com-

Table II. Intrapartum characteristics subsequent to randomization

	Early amniotomy (n = 106)	Late amniotomy (n = 103)
Epidural anesthesia (%)	81.0	76.0
Cervix at epidural (median cm)	3	4
Chorioamnionitis (%)	22.6	6.8*
Uterine hyperstimulation (%)	23.6	15.5
Late decelerations (%)	13.2	16.5
Recurrent moderate to severe variable decelerations (%)	12.3	2.9†
Meconium in labor (%)	23.6	14.6
Cesarean delivery (%)	22.6	22.3

* $p = 0.002$.

† $p = 0.017$.

Table III. Labor characteristics of patients delivered vaginally

	30 min increments		60 min increments	
	Early amniotomy (n = 41)	Late amniotomy (n = 43)	Early amniotomy (n = 41)	Late amniotomy (n = 37)
Time to ROM (hr)	1.5 (4.8)	11.9 (7.8)*	2.1 (6.3)	10.9 (7.1)*
ROM to delivery (hr)	9.7 (5.4)	5.7 (6.4)†	13.2 (7.7)	7.1 (4.8)*
Time to active phase (hr)	6.6 (5.9)	10.9 (7.0)†	10.6 (9.6)	12.4 (7.8)
Time to complete dilatation (hr)	10.1 (6.6)	16.6 (8.0)*	14.7 (10.7)	17.2 (8.3)
Time to delivery (hr)	11.2 (6.7)	17.7 (8.3)*	15.3 (10.7)	18.0 (8.5)

Results expressed as mean $\pm$ SD. ROM, Rupture of membranes.

* $p < 0.001$.

† $p < 0.01$.

monly with early amniotomy in both the 60-minute (19.6% vs 6.4%, $p = 0.08$) and 30-minute (4.0% vs 0%, $p = 0.22$) subgroups, reaching statistical significance for the total population ($p = 0.017$). Similarly, early amniotomy was associated with a higher incidence of chorioamnionitis (22.6% vs 6.8%, $p = 0.002$). Subgroup analysis revealed chorioamnionitis to be associated with early amniotomy only when 60-minute increments of oxytocin were used (39.3% vs 10.6%, $p = 0.001$). Oxytocin adjustment at 30-minute intervals was associated with infrequent chorioamnionitis: 4.0% and 3.6% after early and late amniotomy, respectively (not significant). Birth weights were similar in the two groups (3286 ± 651 vs 3278 ± 535 gm, not significant). Similarly, both groups had a similar incidence of 5-minute Apgar scores < 7 (3.8% vs 1.0%, not significant).

There was no difference in the indications for cesarean delivery in the early and late amniotomy groups (Table II). A total of 13 women underwent cesarean section for fetal distress, eight after early amniotomy and five after late amniotomy. Six women in the early amniotomy group with moderate or severe variable decelerations required cesarean section for fetal distress, whereas none in the late amniotomy group required such intervention.

As anticipated, women enrolled in the early am-

niotomy group had a shorter interval from enrollment to membrane rupture (1.8 ± 5.6 vs 11.4 ± 7.5 hours, $p < 0.001$) compared with the late amniotomy group. In addition, a significant prolongation of the interval from amniorrhexis to delivery was noted with early amniotomy (11.4 ± 6.9 vs 6.4 ± 5.8 hours, $p < 0.001$).

Additionally, early amniotomy was associated with a shorter active phase of labor (8.7 ± 8.3 vs 11.6 ± 7.4 hours, $p = 0.03$), a shorter time to complete dilatation (12.4 ± 9.1 vs 16.9 ± 8.1 hours, $p = 0.001$), and a shorter time to delivery (13.3 ± 9.1 vs 17.8 ± 8.3 hours, $p = 0.001$) compared with late amniotomy. Information regarding the duration of labor in women delivered vaginally is presented in Table III. Subgroup analysis demonstrates a statistically significant shortening of labor only with early amniotomy where oxytocin adjustments were made every 30 minutes. In contrast, there was no such correlation when oxytocin infusion was adjusted every 60 minutes.

Comment

Amniotomy is frequently performed as an adjunct to labor induction and augmentation and during spontaneous labor. Amniorrhexis offers the opportunity for examination of amniotic fluid and for direct measurement of FHR and intrauterine pressure as indicated.

Alternatively, amniotomy may have detrimental effects for both the mother and her fetus, such as an increased frequency of intrapartum infection and an increased incidence of fetal distress from umbilical cord compression or prolapse. However, it remains controversial whether amniotomy provides a favorable risk to benefit ratio, particularly in the setting of labor induction.

Stewart et al.⁹ studied the effect of artificial rupture of membranes in 68 patients in early spontaneous labor and reported that early amniotomy significantly shortened the length of labor and reduced the need for augmentation and instrumental delivery. Similarly, Seitchik et al.¹⁰ reported that amniotomy in the first stage of labor appeared to enhance the cervical dilatation rate in women with advanced cervical dilatation but not in those with less advanced cervical dilatation. In contrast, Fraser et al.¹¹ failed to identify a reduction in labor duration when early amniotomy was performed in spontaneously laboring women.

Unfortunately, there is less information regarding the utility of amniotomy in the setting of labor induction. Bakos and Backstrom¹² published a series of 223 women who were randomized to induction with either oxytocin infusion or amniotomy. Patients were reassessed after 4 hours of induction at which time amniotomy was performed or oxytocin infusion was initiated if labor progress was not deemed adequate. The authors found that amniotomy was associated with a shorter induction to delivery interval. Conversely, the oxytocin group had a higher frequency of postpartum hemorrhage, cervical rupture, and perineal damage. However, they found no statistically significant difference between the groups regarding the rates of cesarean or forceps delivery or fetal morbidity. This study has subsequently been criticized for a selection bias both at entry and during execution of the study.¹³ A subsequent multicenter trial found early amniotomy to be associated with a shorter duration of labor but not a decrease in the incidence of cesarean section.¹¹

In this prospective randomized trial of early versus late amniotomy in labor induction, we found early amniotomy to be associated with a statistically significant 4.5-hour shortening of time to delivery when performed in conjunction with continuous oxytocin infusion. This finding was more pronounced when oxytocin was adjusted at intervals of 30 minutes, a 6.5-hour reduction in this group. Interestingly, such benefit did not appear to be accrued in those women undergoing oxytocin adjustments at 60-minute increments. Alternatively, early amniotomy was associated with a 3.3-fold increase in intrapartum intrauterine infection. In contrast to labor duration, the infection risk associated with early amniotomy was seen only in those women in the 60-minute group. Although it is tempting to claim that women receiving 60-minute oxytocin adjustments are at

increased risk of intrauterine infection compared with those undergoing 30-minute increments, this claim cannot be made because the two arms were not concurrently studied. Finally, early amniotomy was associated with more frequent FHR patterns consistent with significant umbilical cord compression. This finding, likely attributable to prolonged leakage, was not apparent in the 30-minute increment subgroup, but a similar trend was identified among those receiving increments at 60-minute intervals (19.6% vs 6.4%, $p = 0.08$).

Although some care givers prefer to have a single standardized protocol for induction of labor for all patients, it is unlikely that such an approach represents an "optimal" practice. We suggest that labor induction should be individualized on the basis of the indications for induction, presence of risk factors for infection, and anticipated fetal reserve. Some fetuses, such as those with intrauterine growth retardation or uteroplacental insufficiency, will have less reserve and may not tolerate frequent increments in the oxytocin infusion. Therefore judicious oxytocin administration, allowing for observation of fetal response at a plateau of uterine activity, could be achieved by increasing the oxytocin infusion rate at intervals of 60 minutes, as we have described.³ In this setting amniotomy should be performed late in labor to reduce the risk of chorioamnionitis and the risk of significant umbilical cord compression. On the other hand, early amniotomy may be indicated in certain obstetric circumstances so that the amniotic fluid character can be evaluated or so that fetal scalp blood sampling, internal FHR monitoring, or amnioinfusion can be instituted. These patients may benefit from an oxytocin protocol that includes increments at 30-minute intervals. Such a protocol would reduce the risk of both intrauterine infection and umbilical cord compression.

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The accuracy and patient preference for self-collected group B *Streptococcus* cultures

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OBJECTIVE: Our purpose was to determine the accuracy of and patient attitudes regarding self-collected group B *Streptococcus* cultures.

STUDY DESIGN: Women seen for prenatal care at 24 to 42 weeks' gestation were asked to collect distal vaginal and anal samples for group B *Streptococcus*. Subsequently, distal vaginal and anal samples were obtained by the nurse. The patients were then asked their preference toward self-sampling.

RESULTS: A total of 251 women participated in the study. The incidence of positive group B *Streptococcus* cultures was 12.7%, 9.6%, 10.0%, and 7.6% for the patient-collected vaginal and anal and nurse-collected vaginal and anal specimens, respectively. The incidence of group B *Streptococcus* carriage was 17.5% and 13.5% for any positive patient- or provider-collected specimens, respectively, and 19.1% for any positive culture. Single patient-collected vaginal and anal and nurse-collected vaginal and anal samples were insensitive for group B *Streptococcus* carriage (67%, 50%, 52%, 40%, respectively). The combination of patient-collected samples was more sensitive than nurse-collected samples (sensitivity 91.7% vs 70.8%, $p < 0.05$). Repeat sampling of the vagina or anal canal did not offer significant additional benefit to a single culture. Overall, patient-collected samples were 98.4% accurate in predicting group B *Streptococcus* carriage versus 94.4% for nurses. A total of 58% of women preferred obtaining their own specimens, whereas 9.6% found the technique difficult. Ninety percent desired the option of self-sampling in the future.

CONCLUSIONS: Single vaginal or anal cultures were insensitive in detecting group B *Streptococcus* carriage. Combined patient-collected cultures were more sensitive than provider-collected specimens. On the basis of accuracy and patient preference, women should be given the opportunity of combined vaginal-anal self-sampling for group B *Streptococcus* when indicated. (*AM J OBSTET GYNECOL* 1995;173:1325-8.)

Key words: Group B *Streptococcus*, screening, culture, pregnancy

Group B *Streptococcus* is a commensal organism that may, under selected circumstances, cause serious infec-

tion in the gravid woman, her fetus, and newborn infant. A large body of clinical research has focused on identifying risk factors for neonatal group B streptococcal infection and on the reduction of vertical transmission from mother to infant during labor. The American College of Obstetricians has suggested screening of gravid women with risk factors for neonatal sepsis followed by intrapartum treatment of carriers.^{1, 2} Alternatively, the American Academy of Pediatrics has suggested routine screening of all gravid women, again

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